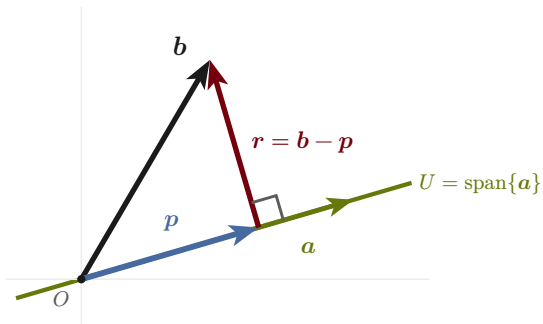


Orthogonal projection onto a subspace

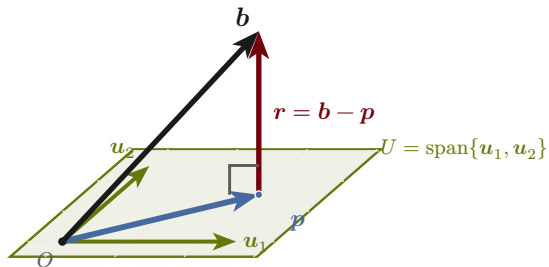
onto a line

$$p = \frac{\langle a, b \rangle}{\langle a, a \rangle} a$$



onto a plane

$$r \perp u_1, r \perp u_2$$



$\rightarrow b$ — original vector $\rightarrow p = \text{proj}_U(b) \in U$ — nearest point $\rightarrow r = b - p \perp U$ — residual

defining property: the residual is orthogonal to the whole subspace, so p minimizes $\|b - x\|$ over $x \in U$ — the basis of least squares and PCA.